

Prasita Dutta

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PROFESSIONAL SUMMARY

Sustainability-focused PhD researcher in Chemistry with a strong foundation in bio-catalysis, spectro-electrochemistry, carbon capture mechanisms, and sustainable energy technologies, aiming to combine scientific expertise with strategic thinking to bridge the gap between research and real-world application. Experienced in leading academic initiatives and communicating complex ideas across diverse teams. Driven by a passion to support organisations in delivering tangible progress towards their net-zero objectives.

EDUCATION

St. Stephen's College, University of Delhi

Bachelor of Science (Chemistry)

Jul. 2022

Delhi, India

➤ **CGPA: 9.51/10.0**; Graduated with distinction.

University of Oxford

PhD in Inorganic Chemistry

Present

Oxford, UK

- **Research focus:** Demystifying complex bio-catalytic procedures in anaerobic organisms to inspire novel carbon-capture technologies and sustainable energy systems through structural and electrochemical manipulations.
- **Key Skills:** Bio-catalysis, Heterogenous catalysis, Electrochemistry, Energy systems, Carbon capture and storage, Hydrogen energy, Spectroscopy, X-Ray crystallography and Green-synthesis.

WORK EXPERIENCE

Chemistry Teaching Laboratory, University of Oxford

Junior Lab Demonstrator (Part-time)

Jan. 2023 – Present

Oxford, UK

- Support experimental design, data analysis and safe laboratory practices in the UK undergraduate curriculum and evaluate undergraduate lab reports and help through troubleshooting complex problems.

Generating Genius

Academic Mentor (Part-time)

Sep. 2024 – Present

Oxford/London, UK

- Lead an energy-themed poster competition promoting awareness of sustainable innovation in the field of chemistry and chemical engineering. Additionally, mentored the students through their Oxbridge admission procedures.

Oxford Cambridge Alumni Tours

Head Tour Guide (Part-time)

Jan. 2023 – Present

Oxford, UK

- As a head guide, led and trained new tour guides in Oxford, showcasing ability to mentor, manage teams, and deliver engaging presentations to diverse global audiences.

FrenchBIC summer school for studying metals in biology

Selected participant

Nov. 2024

Grenoble, France

- Conducted electrochemistry experiments in the University of Grenoble laboratories (in collaboration with Orignalys), and spectroscopic experiments at the European synchrotron (ESRF).

Oxford Global

Chemistry Stream Leader (Voluntary)

Jul. 2023 – Jul. 2024

Oxford, UK and Shanghai, China

- Designed and delivered a course on the applications of green chemistry and sustainability in modern-day research and their industrial implementations at the Oxford Global Summit of Young Leaders in Shanghai. Encouraged a sustainability driven thinking in high-school kids interested in a career in STEM.

WOMENForINDIA Foundation**Jun. 2020 – Jul. 2020***Intern**India*

- Researched and wrote articles on corporate sustainability and circular economy adaptations in Indian industries. Performance based recognition as one of the top 3 interns owing to excellence in research skills and communication.

Homi Bhabha Institute of Science Education**Dec. 2019***Research Scholar**India*

- Treatment of waste-water generated in Indigo-dyeing industries to render them suitable for domestic use. Selected as one of the 50 participants across India to participate in this program.

POSITIONS OF RESPONSIBILITY

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| ➤ Department of Chemistry, Oxford: Supervisor to Master's student | Oct. 2024 – Oct. 2025 |
| ➤ Oxford Cambridge Alumni Tours: Head Guide | Nov. 2024 – Present |
| ➤ Chemistry Joint Consultative Committee: Inorganic Chemistry Representative | Oct. 2023 – Oct. 2024 |
| ➤ Oxford Women in Chemistry: Secretary | Oct. 2023 – Oct. 2024 |
| ➤ Oriel College Middle Common Room (MCR): Secretary | Oct. 2023 – Oct. 2024 |
| ➤ Choreography Society, St. Stephen's College: Administrative Head | Jul. 2020 – Jul. 2021 |

CERTIFICATIONS, SKILLS, INTERESTS & LANGUAGES

- **Certifications:** Associate Fellowship of the Higher Education Academy; Research Integrity Training, Intellectual Property Rights training; Laboratory Health & Safety training.
- **Technical skills:** Microsoft Office suite; Programming languages (Java, Python); Data analysis software (Origin, MATLAB).
- **Soft Skills:** Strategic Thinking; Time management; Team building and communication; Mentoring and training.
- **Languages:** English (fluent); Bengali (native); Hindi (fluent); Spanish (beginner).